

Satwik Pani

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EDUCATION

University of Minnesota Twin Cities, Minneapolis, Minnesota

August, 2021 - October, 2025

Masters of Science, Medical Physics

CAMPEP Accredited

GPA: 3.5

University of Illinois at Urbana-Champaign, Urbana, Illinois

August 2018- December 2020

Bachelors of Science, Physics

GPA: 3.67

EXPERIENCE

Clinical Physics Assistant I

St. Louis, Missouri

Washington University in St. Louis

2023 - Current

- Independently perform machine QA for a Varian Truebeam
- Assisted in performing Annual QA for a Varian Truebeam including recommendations from TG-142 and TG-51.
- Generate patient specific QA for IMRT and SBRT photon treatment plans with Eclipse including ion chamber and portal dosimetry for photon.
- Generated patient specific QA for proton treatment plans with Raystation, including for ion chamber, surface dose plane, and depth dose plane measurements.
- Develop a web-based chart checking script for proton therapy using the Raystation Python API.
- Investigate methods of 3D surface scanning for clinical use using LIDAR and laser scanning.
- Automate simulated phantom generation and CT generation using Python and the XCAT Phantom.
- Built Excel tools to streamline creation of Patient Specific QA task emails.
- Mentored project to rewrite an excel generator for Patient Specific QA in Python.
- Assisted in commissioning of a Varian Halcyon linear
- Contribute to open source software, including pylinac and the Core Imaging Library.
- Automated output factor measurements using Sun Nuclear PC Electrometer, reducing the measurement time from 2 hours to 40 minutes.
- Generated dicoms for automated output factor measurements for both Halycon and Truebeams using pydicom.
- CT Monthly
- Performed data collection for multiple research projects on the novel Varian Accela
 - Developed Python tools to quickly generate dicoms and XMLs for this machine.
 - Generating Varian Developer Mode XML files to collect high sample Winston Lutz measurements, reducing time from 4 hours to 1 hour per collection.
 - Improved efficiency of DLG measurements using the SNC PC Electrometer's logging functionality.
 - Performed commissioning for 2 Varian Accelas, finding suggested improvements and differences from Truebeam.
 - Collected a large set of beam scanning data using a large number of detectors.
 - Characterization of ion chamber recombination factors over multiple detectors due to a higher dose per pulse.
 - Performed daily sets of tests using MaximQA.
 - Performed SRS QA using the SNC Mapcheck, Film, and the W2 detector for dosimetric evaluation of Neo Arc plans.
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Research Assistant/Graduate Assistant

Minneapolis, Minnesota

University of Minnesota Twin Cities

2021-2023

- Designed and iterated on a printed circuit board for hall effect sensors using KiCad.
- Implemented an algorithm in Python to measure motion of a magnet using hall effect sensors in real time using SciPy.
- Verified accuracy of the system by interfacing with a 3D printer to move the magnet.
- Performed quality assurance for diagnostic machines, including X-ray and Fluoroscopy.
- Prepared monthly reports and schedules for diagnostic imaging quality assurance.

Research Assistant

Urbana, Illinois

University of Illinois at Urbana-Champaign

2020-2021

- Simulated electronic responses for many configurations of an silicon photomultiplier(SiPM) array to add Pulse Shape Discrimination functionality to a commercially available board using Ngspice and Python.
- Collected data from a JSeries SiPM to compare to simulation.

PUBLICATIONS

1. Pani, Satwik, Brandon Nguyen, Yoichi Watanabe, and Damien Matthew. 2024. "Preliminary Evaluation of Hall Effect Sensor Array for Patient Motion Tracking." <https://doi.org/10.13140/RG.2.2.29841.24169>
2. Leland, John, Ming Fang, Satwik Pani, Yuri Venturini, Marco Locatelli, and Angela Di Fulvio. 2024. "Enabling Pulse Shape Discrimination with Commercial Asics." *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 1064 : 169438. <https://doi.org/10.1016/j.nima.2024.169438>

SKILLS

Programming: Python, Rust, Embedded, Sensors

Clinical: Truebeam, Halcyon, Eclipse, Raystation, Doselab, PTW Beamscan, PTW Starcheck, SNC PC Electrometer, SNC 1D Scanner, SNC SRS Mapcheck, Patient Specific QA, Linear Accelerator QA, Fluoroscopic QA, X-Ray QA

Software: Excel, Microsoft Power Platform